

Financial Econometrics: Basics of Time Series Analysis 1

Sora Chon

Inha University

Note: This lecture note covers Ch 2,3,4,8,15 of Introductory Econometrics: A Modern Approach (6th edition) by Jeffrey M. Wooldridge.

Motivation

Stationarity v.s. Non-stationarity

Basic Time Series Models

- Static Models

- Finite Distributed Lag (FDL) Models

- Models with Lagged Dependent Variables

- Deterministic Trend

- Seasonality

Finite-Sample Analysis of OLS for Time Series Data

Outline

Motivation

Stationarity v.s. Non-stationarity

Basic Time Series Models

- Static Models

- Finite Distributed Lag (FDL) Models

- Models with Lagged Dependent Variables

- Deterministic Trend

- Seasonality

Finite-Sample Analysis of OLS for Time Series Data

Differences Between Cross-Sectional and Time Series Data

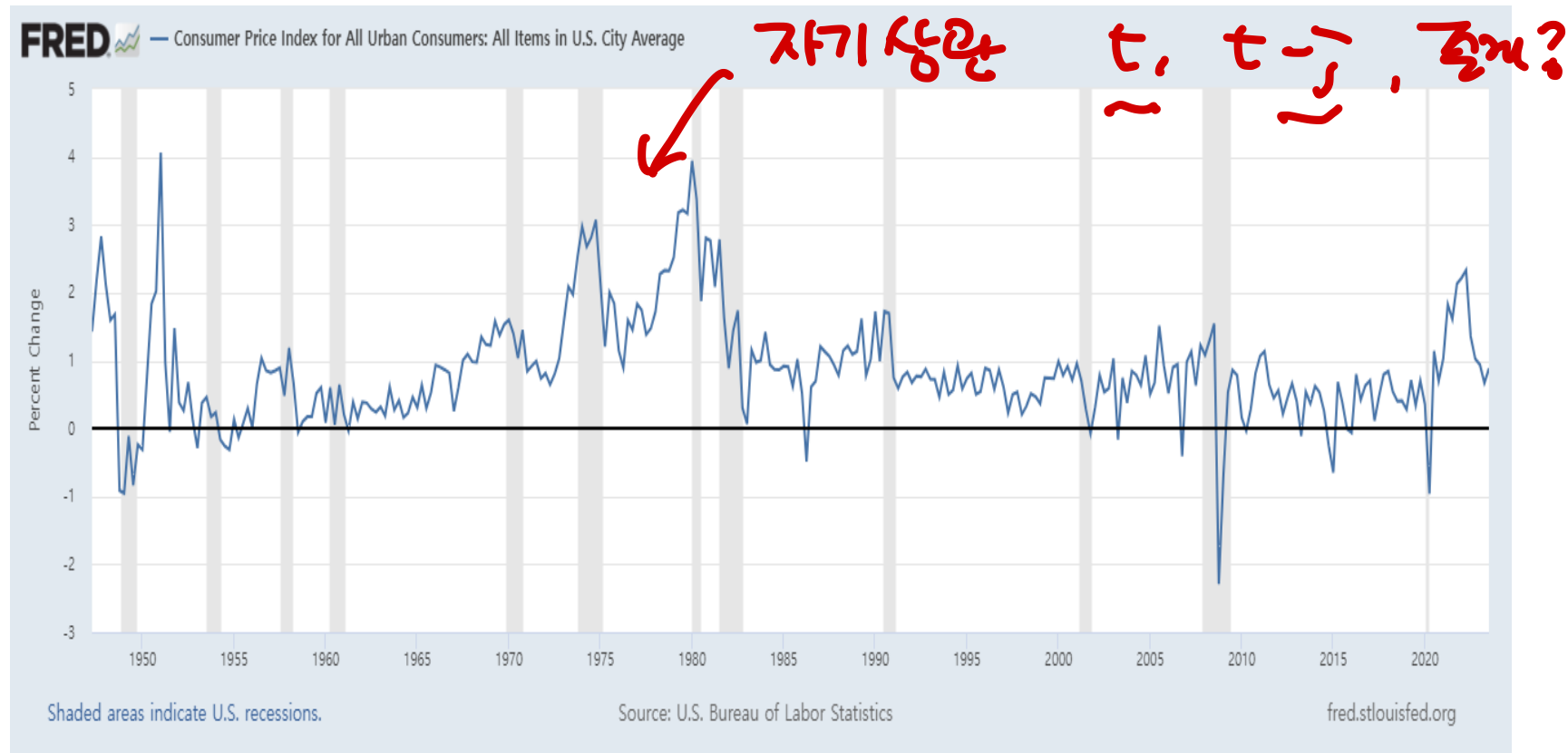


Figure: U.S. Inflation Rate

- ▶ Random sampling cannot be applied to time series variables. ~~X~~
- ▶ The order of time series observations matters a lot. ✓
- ▶ Time series observations are serially correlated. ✓

Differences Between Cross-Sectional and Time Series Data

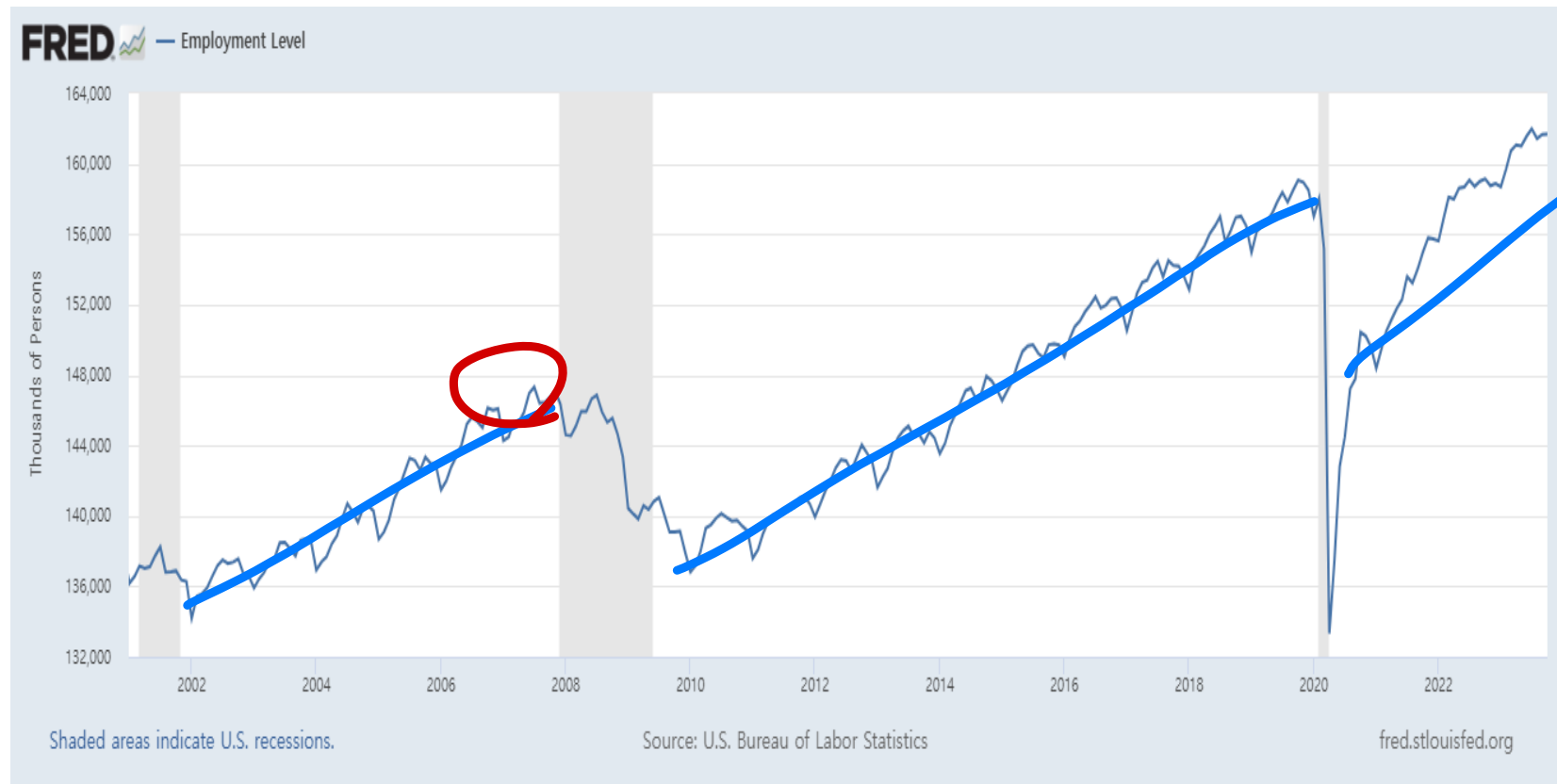


Figure: U.S. Employment

- ▶ Exhibit time trend.
- ▶ Exhibit seasonality.

Differences Between Cross-Sectional and Time Series Data

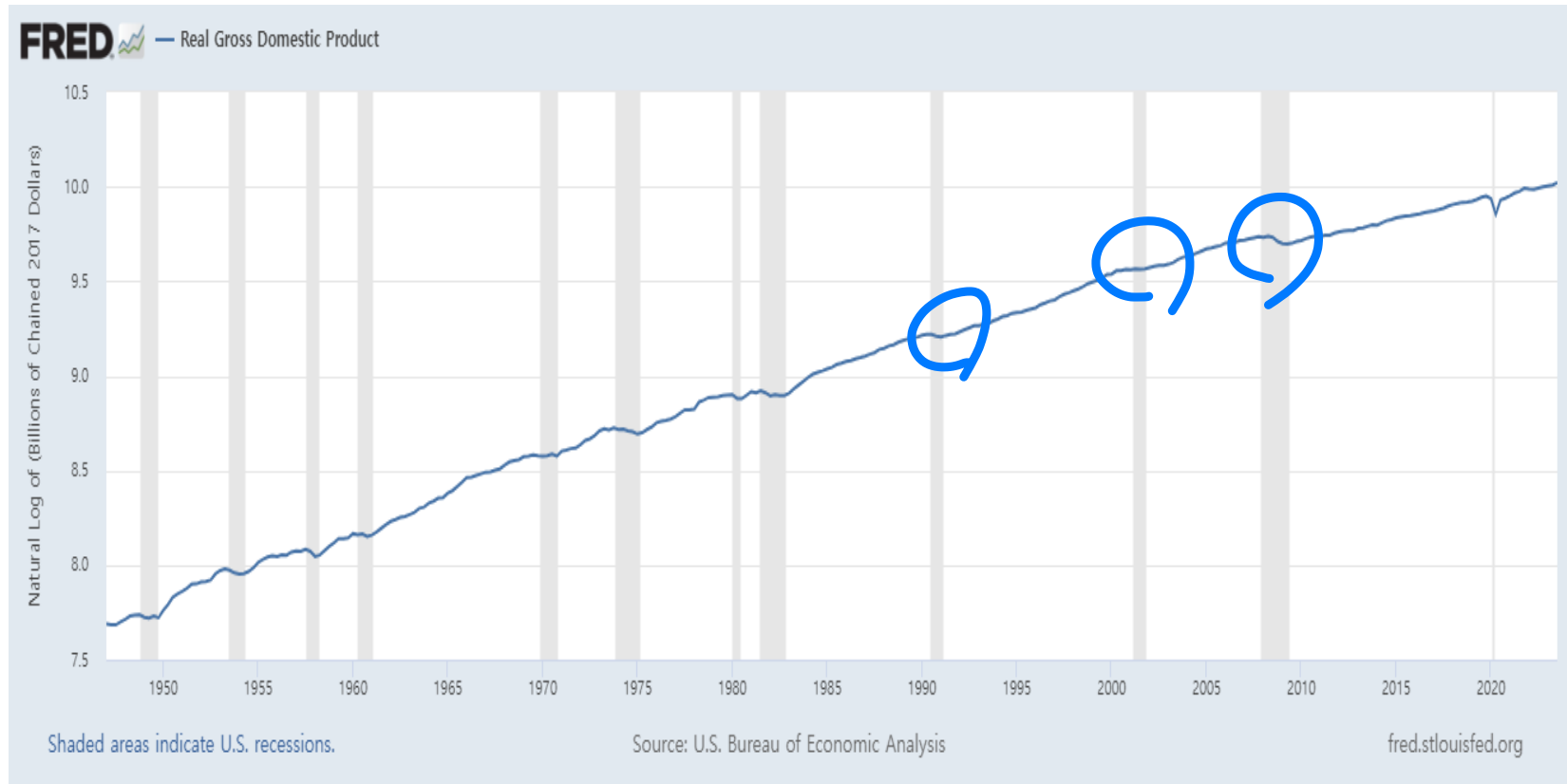


Figure: U.S. (log) Real GDP (seasonally adjusted)

- ▶ Exhibit time trend.
- ▶ Government agencies often provide seasonally adjusted time series observations.

Differences Between Cross-Sectional and Time Series Data

1. Time series data come with a temporal ordering, usually from earliest to latest. For many purposes, the ordering of the data is important.

→ 2. We cannot think of time series data as a random sample of units (individuals, firms, schools, and so on) from a large population. Therefore, we cannot realistically impose random sampling.

✗ 3. Because random sampling implies the observations are independent of one another, we did not have to worry about correlation across observations in the cross-sectional case. But time series data are almost always correlated across time, sometimes very strongly.

Differences Between Cross-Sectional and Time Series Data

↗️ **부사는 Big.**

확인? ✓

4. Very strong time series correlation (persistence) can cause problems for the usual ordinary least squares inference

5. Many time series data sets exhibit trends – sometimes up, sometimes down, and sometimes more subtle. It is easy to think two trending series are “related” to one another when they are both growing or shrinking but for different reasons. We will cover simple ways of controlling for trends in regression.

6. For time series data at the monthly and quarterly frequencies, seasonality can be an issue. We will see simple methods of allowing for seasonality in OLS.

Outline

Motivation

Stationarity v.s. Non-stationarity

Basic Time Series Models

- Static Models

- Finite Distributed Lag (FDL) Models

- Models with Lagged Dependent Variables

- Deterministic Trend

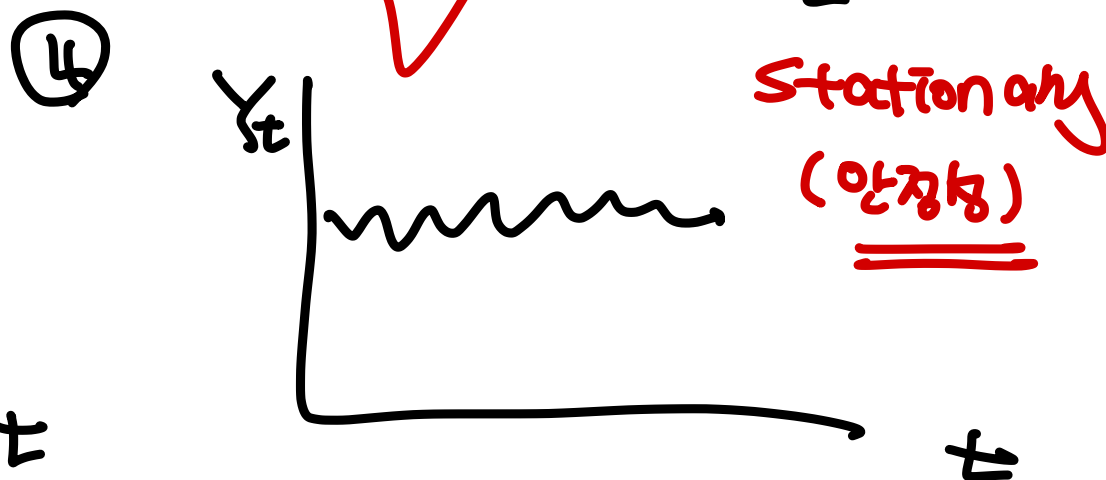
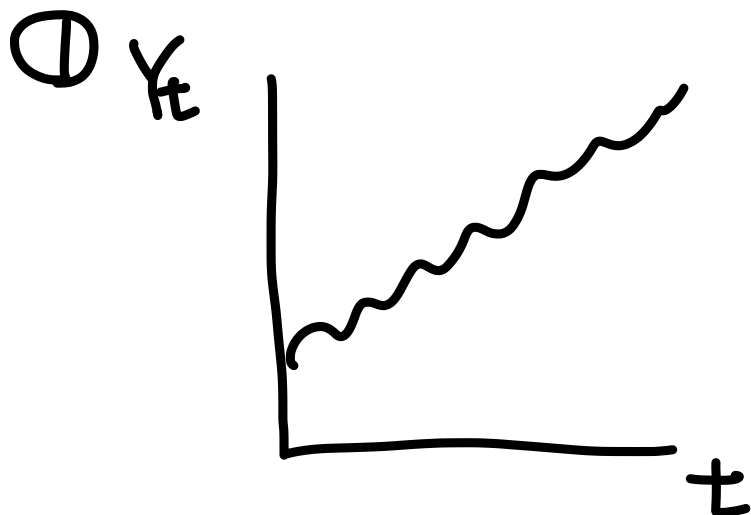
- Seasonality

Finite-Sample Analysis of OLS for Time Series Data

Ergodicity, strict, and weak stationarity

► Graphically, three conditions are connected as

$\textcircled{Y_t}$



Intuition: $\{Y_t\}$ is stationary if all aspects of its behavior are unchanged by shifts in time

✓

- L

- ◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ▶ ↺ 🔍 ↻

Weak stationary process

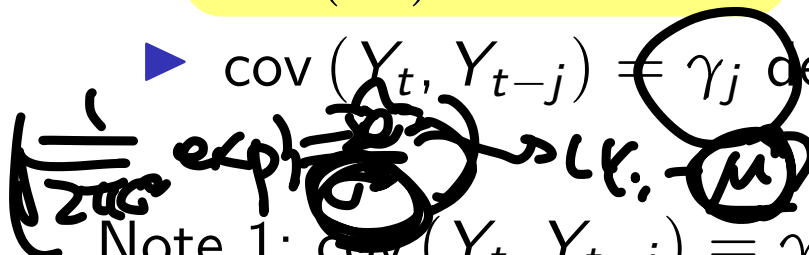


④



Covariance (Weakly) Stationary Processes $\{Y_t\}$:

- ▶ $E[Y_t] = \mu$ for all t
- ▶ $\text{var}(Y_t) = \sigma^2$ for all t
- ▶ $\text{cov}(Y_t, Y_{t-j}) = \gamma_j$ depends on j and not on t



$\lim_{j \rightarrow \infty} \gamma_j = 0$

Note 1: $\text{cov}(Y_t, Y_{t-j}) = \gamma_j$ is called the j -lag auto-covariance and measures the direction of linear time dependence

Note 2: A stationary process is covariance stationary if $\text{var}(Y_t) < \infty$ and $\text{cov}(Y_t, Y_{t-j}) < \infty$

* Stochastic process vs. time-series Data

(ARMA I. // 예측가능) (i) : 통계적 인과관계, 예측?
 $\Delta X_n \rightarrow \Delta Y_n$

(+) 어떤 예측?

Y_{t+j} ?



e_t 충격 (error term at time t)

$\frac{\partial Y_{t+j}}{\partial e_t}$?

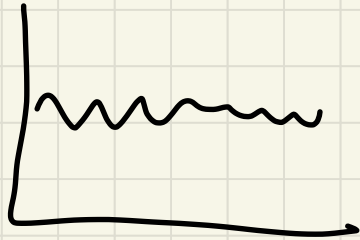
Impulse-Response 충격 - 반응 정도

① transitory effect $\nabla \Rightarrow$ time-sens
 ② permanent effect $\nabla \Rightarrow$ 1

인식 가능 $\rightarrow$ (AR) (AR)
 $\rightarrow$ (MA) $\Rightarrow$ (MA)

이항사건-시간성 확률과정

: 100 일의 기간에 1 번씩 일어날 확률



< 확률적 ex-ante

< Y_1 Y_2 Y_3 Y_4 ... Y_{99} Y_{100} >

$E(Y)$

Stochastic process



a sequence of random variables ordered in time.

(이항사건)

Data
 y_1^1
 y_1^2
 $\vdots$
 y_1^j
 $\vdots$

Data
 y_2^1
 y_2^2
 $\vdots$
 y_2^j
 $\vdots$

...

y_{100}^1
 y_{100}^2
 $\vdots$
 y_{100}^j

이항사건의
확률과정

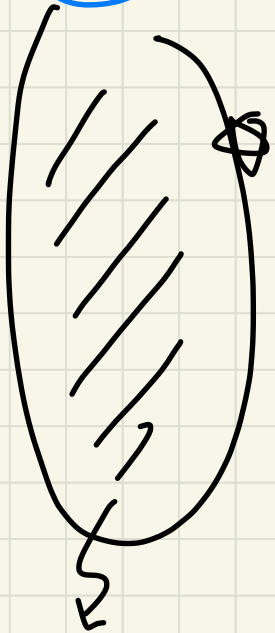
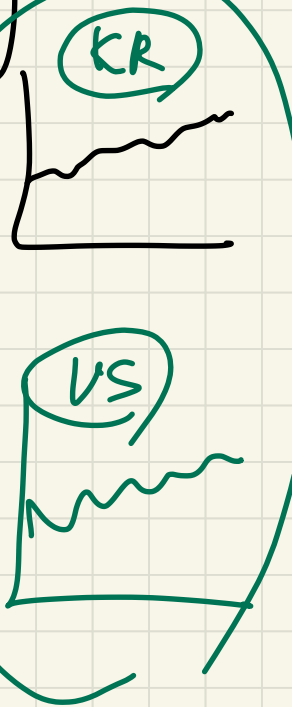
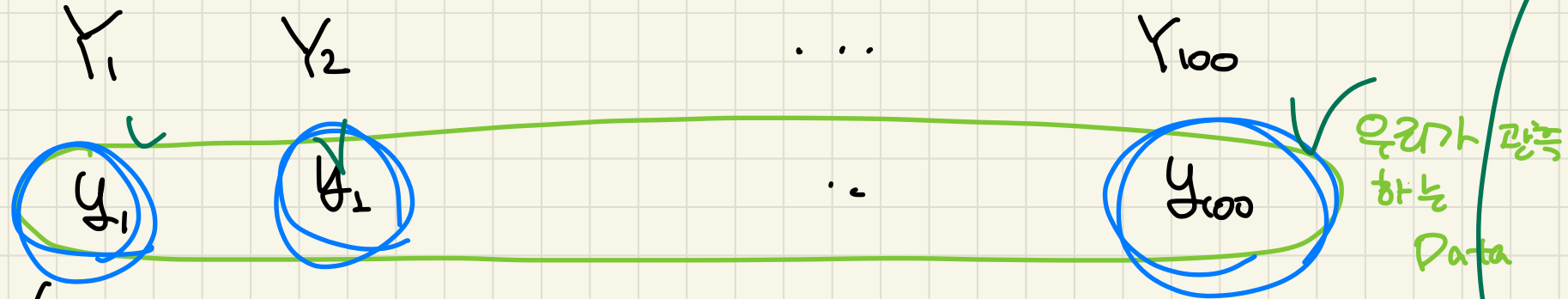
Time-series Data:

A particular realization of stochastic process.

$\Rightarrow$ Ex-post

제1) 한국 GDP. (단위 : 2006 ~ 2015)

경제성장률 (CAGR)

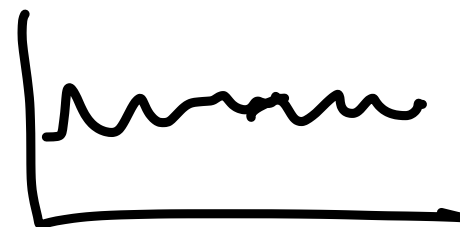


비트 X.

Can't see!

우리가
관측

Strong (strict) stationary process



- A stochastic process $\{Y_t\}_{t=1}^{\infty}$ is strictly stationary if, for any given finite integer r and for any set of subscripts $t_1, t_2, \dots, t_r$ the joint distribution of

$$(Y_{t_1}, Y_{t_2}, \dots, Y_{t_r})$$

depends only on $t_1 - t, t_2 - t, \dots, t_r - t$ but not on t .

- ▶ For example, the distribution of (Y_1, Y_5) is the same as the distribution of (Y_{12}, Y_{16}) .
- ▶ For a strictly stationary process, Y_t has the same mean, variance (moments) for all t .
- ▶ Any function/transformation $g(\cdot)$ of a strictly stationary process, $\{g(Y_t)\}$ is also strictly stationary. E.g., if $\{Y_t\}$ is strictly stationary then $\{Y_t^2\}$ is strictly stationary.

$$f(Y_1, Y_2, \dots, Y_r) = f(Y_{1+r}, Y_{2+r}, \dots, Y_{r+r}) \quad \text{for any } r$$

(Joint pdf)

Ergodicity

[- 중간고사: 4/21 : 화 - 수업시간
- 4/30: 휴강 → 올라임. (5/5)
- (tw)]

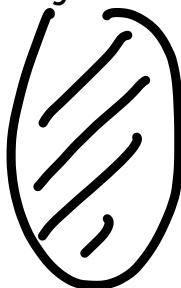
Loosely speaking, a stochastic process $\{Y_t\}_{t=1}^{\infty}$ is ergodic if any two collections of random variables partitioned far apart in the sequence are almost independently distributed. The formal definition of ergodicity is highly technical.(see Hayashi 2000, p. 101 and Hamilton, 1994 page 47).

Econometrics by Hayashi, Fumio (2000)

Time Series Analysis by James Hamilton (1994)

If Y_t is an ergodic process,

$$E[Y_t] = \text{plim} \frac{1}{J} \sum_{j=1}^J Y_t^{(j)} = \text{plim} \frac{1}{T} \sum_{t=1}^T Y_t$$



Ergodicity

1, 2 moments.

Let $\{Y_t\}$ be a covariance stationary process with mean $E[Y_t] = \mu$ and autocovariances $\gamma_j = \text{cov}(Y_t, Y_{t-j})$. If

$$\sum_{j=0}^{\infty} |\gamma_j| < \infty \quad ? \Rightarrow \text{ergodic}$$

then $\{Y_t\}$ is ergodic for the mean. $t \rightarrow t-j$

$j \rightarrow \infty$

→ 이걸 아주 멀리 떨어진 값의 Data 평균을

구하면 된다면 if $\Rightarrow$ stationary.

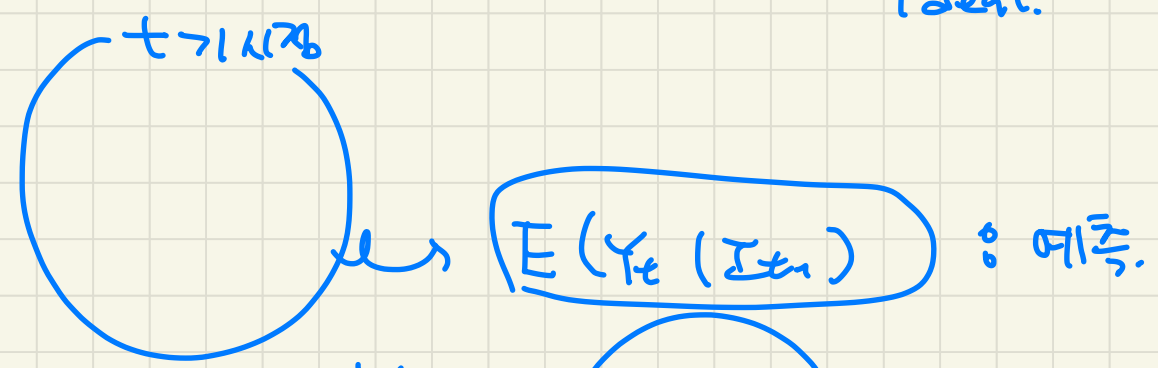
ergodic $\Rightarrow$ stationary

* Note *

- Impulse - Response !! $\frac{\partial Y_{t+j}}{\partial \epsilon_t} = ?$

- $Y_t = f(\underbrace{x_t, z_t, \dots}_{\nearrow \eta_t}) + u_t$

- $E(Y_t | \eta_t) \underset{\text{all factors}}{=} \underline{\underline{E(Y_t | I_t)}}$
Ideal.



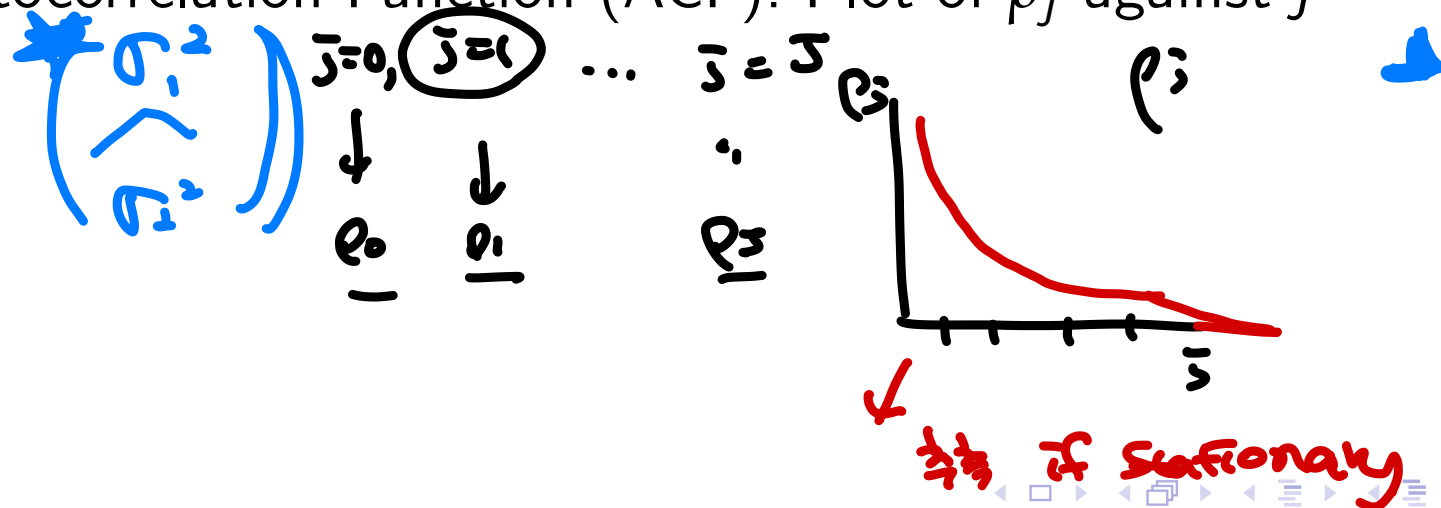
forecast
predict //

$\underline{Y_t} - \underline{\underline{E(Y_t | I_t)}} = \underline{\underline{\text{예측오차.}}}$

Auto-Correlation Function (ACF)

$$\text{corr}(Y_t, Y_{t-j}) = \rho_j = \frac{\text{cov}(Y_t, Y_{t-j})}{\sqrt{\text{var}(Y_t) \text{var}(Y_{t-j})}} = \frac{\gamma_j}{\sigma^2}$$

- $\text{corr}(Y_t, Y_{t-j}) = \rho_j$ is called the j-lag autocorrelation and measures the direction and strength of linear time dependence
- By stationarity $\text{var}(Y_t) = \text{var}(Y_{t-j}) = \sigma^2$.
- Autocorrelation Function (ACF): Plot of ρ_j against j



Dynamic stability condition

To show that Y_t is ergodic in mean, we need to show

$$\sum_{j=0}^{\infty} |\gamma_j| < \infty$$

or

$$\lim_{j \rightarrow \infty} \gamma_j = 0, \text{ or } \lim_{j \rightarrow \infty} \rho_j = 0$$

$$\frac{\partial Y_t}{\partial \theta_t} = ?$$

e.g. > 관측치
정제 / 중요 변수
...
① unit-root
② × 상관.
: stationary
effect

- ▶ $\lim_{j \rightarrow \infty} \rho_j = 0$ is referred to as the dynamic stability condition.
- ▶ We can empirically check if the dynamic stability condition holds given time series data. (unit root tests) Data on unit root
동적 안정성 : permanent effect
- ▶ If the dynamic stability condition holds, the corresponding time series variable is ergodic. So it is stationary.
- ▶ If the time series variable is stationary, we can get meaningful empirical results.

Outline

Motivation

Stationarity v.s. Non-stationarity

Basic Time Series Models

- Static Models
- Finite Distributed Lag (FDL) Models
- Models with Lagged Dependent Variables
- Deterministic Trend
- Seasonality

Finite-Sample Analysis of OLS for Time Series Data

Static Models

- The simplest model with time series is called a **static model**, which related the outcome on the dependent variable at time t , y_t , to one or more explanatory variables dated at the same time.
- With just one explanatory variable z_t , we have

$$\underline{y_t} = \beta_0 + \beta_1 \underline{z_t} + u_t$$

log?

EXAMPLE: Static Phillips Curve

One way to write a static Phillips curve is

Box) (Tied)
→ (u_t)

장기적 MV=PR

U_t ↓ (노동시장이 tight) → Wage ↑ → P_t ↑ → $\frac{\Delta P_t}{P_t}$

$$\checkmark \quad \underline{inf_t} = \beta_0 + \beta_1 \underline{unem_t} + u_t$$

US.
or
미국 인플레이션

where inf_t is, say, the annual rate of inflation during year t , and $unem_t$ is annual unemployment rate during year t . β_1 is supposed the measure the tradeoff between inflation and unemployment.

trade-off

Static Models

- Static models are generally used when we are interested in a contemporaneous relationship. But they cannot capture effects that take place with a lag.

동기관계. ($t+1$)



- Static models are not good for forecasting. For one, they ignore the fact that usually past outcomes on y help predict further values of y . Second, to forecast y_{t+1} at time t we would have to know or forecast z_{t+1} at time t .

FDL Models

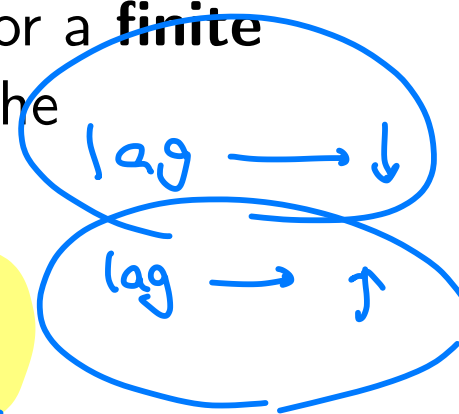
Static model



$$E(u_t | z_t, z_{t-1}, z_{t-2}) = 0$$

- Suppose that we think that a change in a variable, z , today, can affect y up to two periods into the future. This calls for a **finite distributed lag model** with two lags (in addition to the contemporaneous variable):

$$y_t = \alpha_0 + \delta_0 z_t + \delta_1 z_{t-1} + \delta_2 z_{t-2} + u_t$$



- Such models are good for estimating lagged effects of, say, policy.

with $\hat{CPI}_t$, $\frac{12\hat{CPI}_t}{12}$
with GDP_t (Δu)
yearly HCE_t

15/8/29

EXAMPLE: Effects of Personal (Tax) Exemption on Fertility

For purely biological reasons, the effect of making it monetarily more attractive to have children – by increasing the value of the personal exemption, pe , – is unlikely to have a purely contemporaneous effect. Plus, people often react with a lag to policy changes.

- Allowing for a two-year effect on the general fertility rate (gfr):

$$gfr_t = \alpha_0 + \delta_0 pe_t + \delta_1 pe_{t-1} + \delta_2 pe_{t-2} + u_t \quad \checkmark$$

$$\delta_0 + \delta_1 + \delta_2$$

$$\checkmark E(u_t | \text{history}) = 0$$

FDL Models

- An FDL model of order q is

$$y_t = \alpha_0 + \delta_0 z_t + \delta_1 z_{t-1} + \delta_2 z_{t-2} + \dots + \delta_q z_{t-q} + u_t$$

- As a practical matter, the choice of q can be hard. Often dictated by frequency of data. With annual data, q is usually small. With monthly data, q is often chosen as 12 or 24 or even higher, depending on how many months of data we have.
- As we will see, under some assumptions we can use an F test to see if additional lags are jointly significant.

FDL Models

- The coefficient on the contemporaneous z , δ_0 , is called the **impact propensity (IP)**:

$$\text{Impact Propensity} = \delta_0$$

It tells us the immediate change in y when z increases by one unit.

- The sum of all lag coefficients is called the **long run propensity (LRP)**.

$$\text{Long Run Propensity} = \delta_0 + \delta_1 + \dots + \delta_q$$

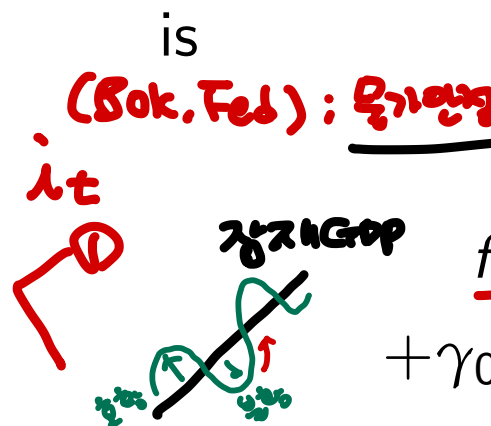
The LRP gives us the answer to the following thought experiment. Suppose the level of z increases *permanently* today. For example, the minimum wage increases by \$1.00 per hour and stays there. The LRP is the (ceteris paribus) change in y after the change in z has passed through all q time periods.

FDL Models

- Notice that if z increases by one unit today, but then falls back to its original level in the next period, the lag distribution tells us how y changes in each future period. Eventually y falls back to its original level with a temporary change in z .
- With permanent change in z , y changes to a new level, and the change from the old to the new level is the LRP.

FDL Models

- We can, of course, have more than one variable appear with multiple lags. For example, a simple equation to explain how the Federal Reserve Bank in the U.S. changes the Federal Funds Rate is



$$\begin{aligned}
 \text{fbrate}_t = & \alpha_0 + \delta_0 \text{inf}_t + \delta_1 \text{inf}_{t-1} + \delta_2 \text{inf}_{t-2} \\
 & + \gamma_0 \text{gdpgap}_t + \gamma_1 \text{gdpgap}_{t-1} + \gamma_2 \text{gdpgap}_{t-2} + u_t,
 \end{aligned}$$

where inf_t is the inflation rate and gdpgap_t is the GDP gap (measured as a percent). The available data are usually quarterly so we probably would try at least four lags.

FDL Models

- FDLs are often more realistic than static models, and they can do better forecasting because they account for some dynamic behavior. But they are not usually the most preferred for forecasting because they do not allow lagged outcomes on y to directly affect current outcomes.

Models with Lagged Dependent Variables

$$\underline{t-j} \quad j=1 \dots$$

- With time series data, there is the possibility of allowing past outcomes on y to affect current y . The simplest model is

$$y_t = \beta_0 + \beta_1 y_{t-1} + u_t,$$

which is a simple regression model for time series data where the explanatory variable at time t is y_{t-1} .

- Called an autoregressive model of order 1, or AR(1).

Models with Lagged Dependent Variables

- This simple model typically does not have much economic or policy interest because we are just using lagged y to explain current y .
- We can add even more lags of y to explain y_t .
- Autoregressive models can be remarkably good at forecasting, even compared with complicated economic models.

Models with Lagged Dependent Variables

i_t (inflation)
 π_t (inflation)
 $unemp_t$ (unemployment)

- It is easy to add other explanatory variables along with a lag. For example,

$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 z_t + u_t$$

- β_2 measures the effect of changing z_t on y_t , holding fixed y_{t-1} . It is a kind of short-run effect of z on y .
- Other extensions can be very useful for forecasting. For example, to forecast inflation one period out, we include only lags of variables:

$$inf_t = \beta_0 + \beta_1 inf_{t-1} + \beta_2 unem_{t-1} + u_t$$

$E(u_t | I_{t-1}) = 0$

- This model implies we forecast next period's inflation, inf_{t+1} , as a linear function of this period's inflation and unemployment:

$$\Rightarrow E(inf_{t+1} | I_t) = \beta_0 + \beta_1 inf_t + \beta_2 unem_t$$

$E(inf_t | I_{t-1}) = \beta_0 + \beta_1 inf_{t-1} + \beta_2 unem_{t-1}$

(We cannot forecast the error term next period, $t + 1$, so it is set to its mean value, zero.)

$inf_{t+1} - E(inf_{t+1} | I_t) = u_{t+1}$

Characterizing Trending Data

$$Y_t \rightarrow \underbrace{\Delta Y_t}_{\text{seasonal}}$$

→ APL, APL, AR... ..
ARMA

- Many series tend to increase over time, at least on average. They typically have up and down periods, but the overall trend is up. (An example is gross domestic product.)
- Other variables tend to decline over time (such as the rate of traffic fatalities).
- Whether a series grows or shrinks over time, care needs to be used because we can find spurious relations among trending variables that have nothing to do with each other.

Characterizing Trending Data

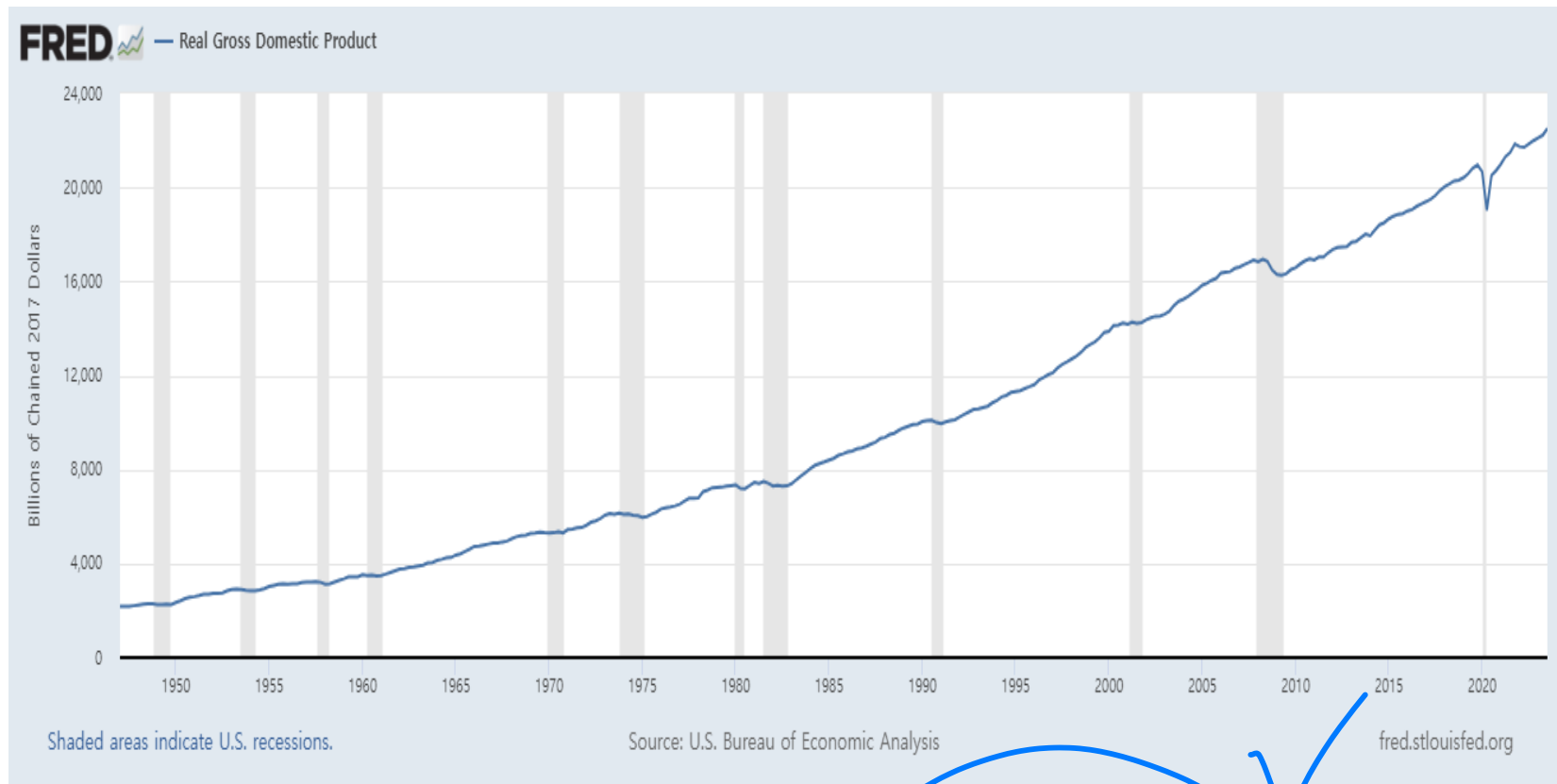


Figure: U.S. Real GDP (seasonally adjusted)

Characterizing Trending Data

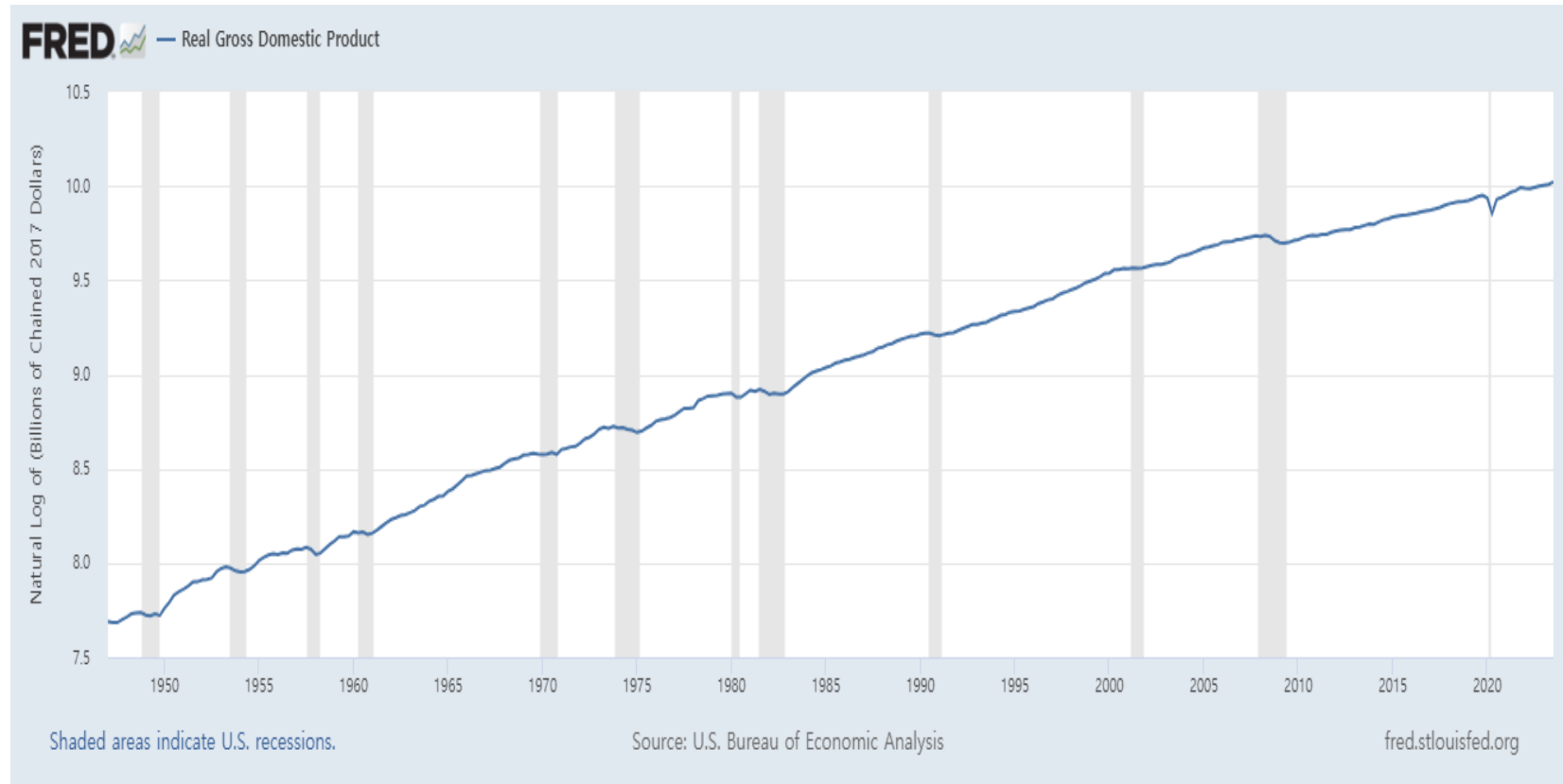


Figure: U.S. (log) Real GDP (seasonally adjusted)

Characterizing Trending Data

- It is very common to use linear time trends. A simple representation is

$$\begin{aligned} y_t &= \alpha_0 + \alpha_1 t + e_t \\ E(e_t) &= 0 \end{aligned}$$

- So the average value of y_t is a linear function of time:

$$E(y_t) = \alpha_0 + \alpha_1 t$$

constant $\times$

and then e_t is the deviations about the trend.

Characterizing Trending Data

- Define the change in y_t from period $t - 1$ to t as

$$\Delta y_t = y_t - y_{t-1}$$

Then under the linear trend representation,

$$\begin{aligned}\Delta y_t = y_t - y_{t-1} &= (\alpha_0 + \alpha_1 t + e_t) - (\alpha_0 + \alpha_1(t-1) + e_{t-1}) \\ &= \alpha_1 + \Delta e_t\end{aligned}$$

Because $E(\Delta e_t) = E(e_t - e_{t-1}) = 0$,

$$\alpha_1 = E(\Delta y_t) \text{ for all } t$$

- In other words, α_1 is the average *change* in y_t over each period.

Characterizing Trending Data

Δ

In

- Other series are better approximated by exponential trends (e.g. real GDP and many other economic variables).

Infl & Unemp

- With exponential trends, the rate of growth is constant.

Characterizing Trending Data

- For strictly positive variables, we can capture an exponential trend as

$$y_t = \exp(\beta_0 + \beta_1 t + e_t),$$

where $E(e_t) = 0$.

- Taking logs gives

$$\boxed{\log(y_t) = \beta_0 + \beta_1 t + e_t} \Rightarrow \Delta(\ln y_t) = \frac{(\ln y_t - \ln y_{t-1})}{1} \leftarrow$$

Stationary!

- In other words, the log of the variable follows a linear trend.

Characterizing Trending Data

- How can we interpret β_1 ? Define the change in the log as

$$\Delta \log(y_t) = \log(y_t) - \log(y_{t-1})$$

- If we set e_t and e_{t-1} to zero, we get

$$\Delta \log(y_t) = \beta_1$$

- Remember that the change in the log approximates the growth rate (as a decimal). Therefore,

$$\beta_1 \approx \frac{(y_t - y_{t-1})}{y_{t-1}}$$

Handwritten notes in blue ink:

Top right: $\beta_1 = ?$

Below it: $y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 e_t$

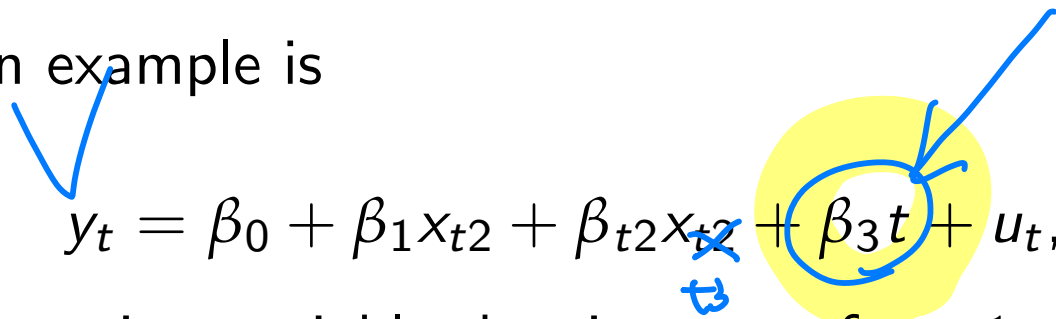
Below that: $+ \alpha_3 e_{t-1}$

Arrows point from the handwritten equations to the printed text "Therefore," and the printed equation $\beta_1 \approx \frac{(y_t - y_{t-1})}{y_{t-1}}$.

Using Trending Data in Regression Analysis

- In order to be sure we are capturing a true relationship between y_t and the explanatory variables when at least one is linearly trending, we can add a linear time trend to the regression.

- An example is


$$y_t = \beta_0 + \beta_1 x_{t2} + \beta_{t2} x_{t2} + \beta_3 t + u_t, \quad t = 1, 2, \dots, n$$

where t is a variable that just runs from 1 to n .

- This equation allows us to control for a linear trend that affects y_t and may also be related to trends in x_{t1} , and x_{t2} .

Using Trending Data in Regression Analysis



- $\rightarrow$ β_3
- Leaving out t causes bias – perhaps severe bias – in estimating β_1 and β_2 .
 - Under our classical model assumptions, we can use the usual statistics, and confidence intervals, in the usual way. This includes for testing the coefficient on the trend, β_3 .

Seasonality

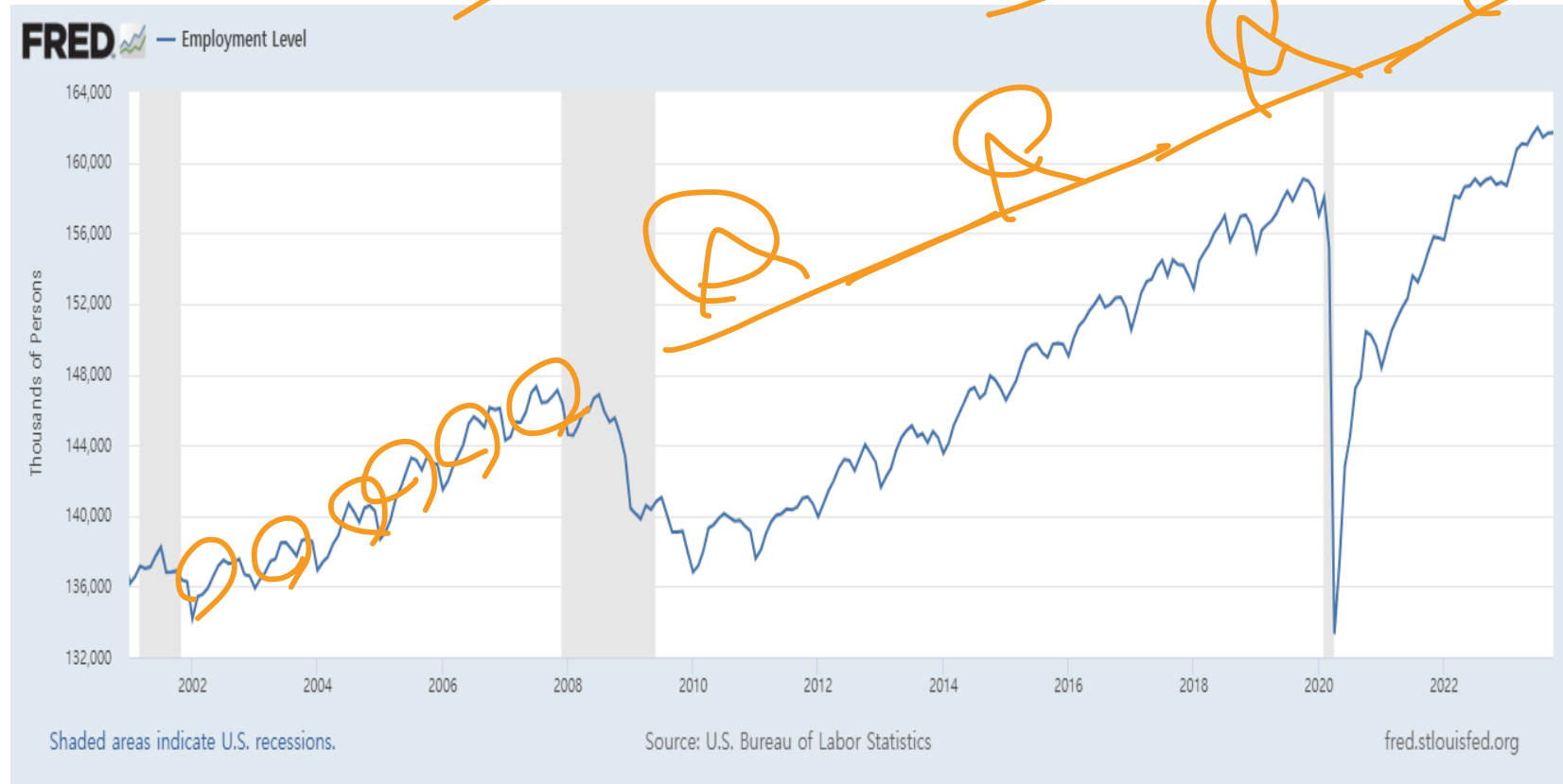


Figure: U.S. Employment

Seasonality

- When the data are quarterly or monthly (and, less often, weekly or daily), we may need to account for different seasonal patterns in y_t or the x_{tj} .
- Often quarterly or monthly data are seasonally adjusted by the government, in which case we can ignore seasonality. But in other cases the data have not been adjusted.
- An easy way to account for seasonality is to include **seasonal dummy variables** in the regression equation. We choose a quarter (usually the first) or a month (usually January) and then include dummies for the remaining quarters or months.

Seasonality

- For example, if we have monthly data, we define dummies feb_t , mar_t , ..., nov_t , dec_t , equal to one when t corresponds to the appropriate month, and zero otherwise.

$$y_t = \beta_0 + \delta_1 feb_t + \delta_2 mar_t + \dots + \delta_{11} dec_t + \beta_1 x_{t1} + \dots + \beta_k x_{tk} + u_t$$

- We can include trends along with seasonal dummies, too.
- Under our assumptions, we can use a joint F test for whether the seasonal dummies are jointly significant.

Outline

Motivation

Stationarity v.s. Non-stationarity

Basic Time Series Models

Static Models

Finite Distributed Lag (FDL) Models

Models with Lagged Dependent Variables

Deterministic Trend → trend ଅନିବାର୍ଯ୍ୟ

Seasonality

✓ Finite-Sample Analysis of OLS for Time Series Data

Motivation

- What kinds of assumptions do we need to get meaningful empirical results from time series observations when employing OLS?

Assumptions

Assumption TS.1 (Linear in Parameters): For a time series process $\{(y_t, x_{t1}, \dots, x_{tk}) : t = 1, \dots, T\}$,

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \dots + \beta_k x_{tk} + u_t, \quad t = 1, \dots, T.$$

- We explicitly use t to index time because we cannot sensibly think of an underlying population.

Assumptions

Assumption TS.2 (No Perfect Collinearity): Each x_{tj} varies somewhat over time, and no explanatory variable is an exact linear function of the others.

- As before, TS.2 rules out *perfect* correlation among independent variables. The consequences of high correlation among the x_{tj} is the same as the cross section (CS) case: it is not a violation of any assumptions, but it can cause difficulty in precisely estimating parameters.

Identify? . Variance ↑

가변성 ↑

무의한
양상

Assumptions

Repeated sampling .

past. current
future

Assumption TS 3 (Zero Conditional Mean): For each t ,

$$E(u_t | \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_t, \dots, \mathbf{x}_T) = 0$$

where $\mathbf{x}_t = (x_{t1}, \dots, x_{tk})$ is the collection of explanatory variables at time t .

- In practice, we ask whether u_t is uncorrelated with each x_{sj} for all t and s , including $t = s$ and all variables $j = 1, \dots, k$,

"consistent"

~~Assumption TS.3 is often called **strict exogeneity** of $\{\mathbf{x}_t : t = 1, \dots, T\}$.~~

bias $\rightarrow 0$

$\neq 0$,

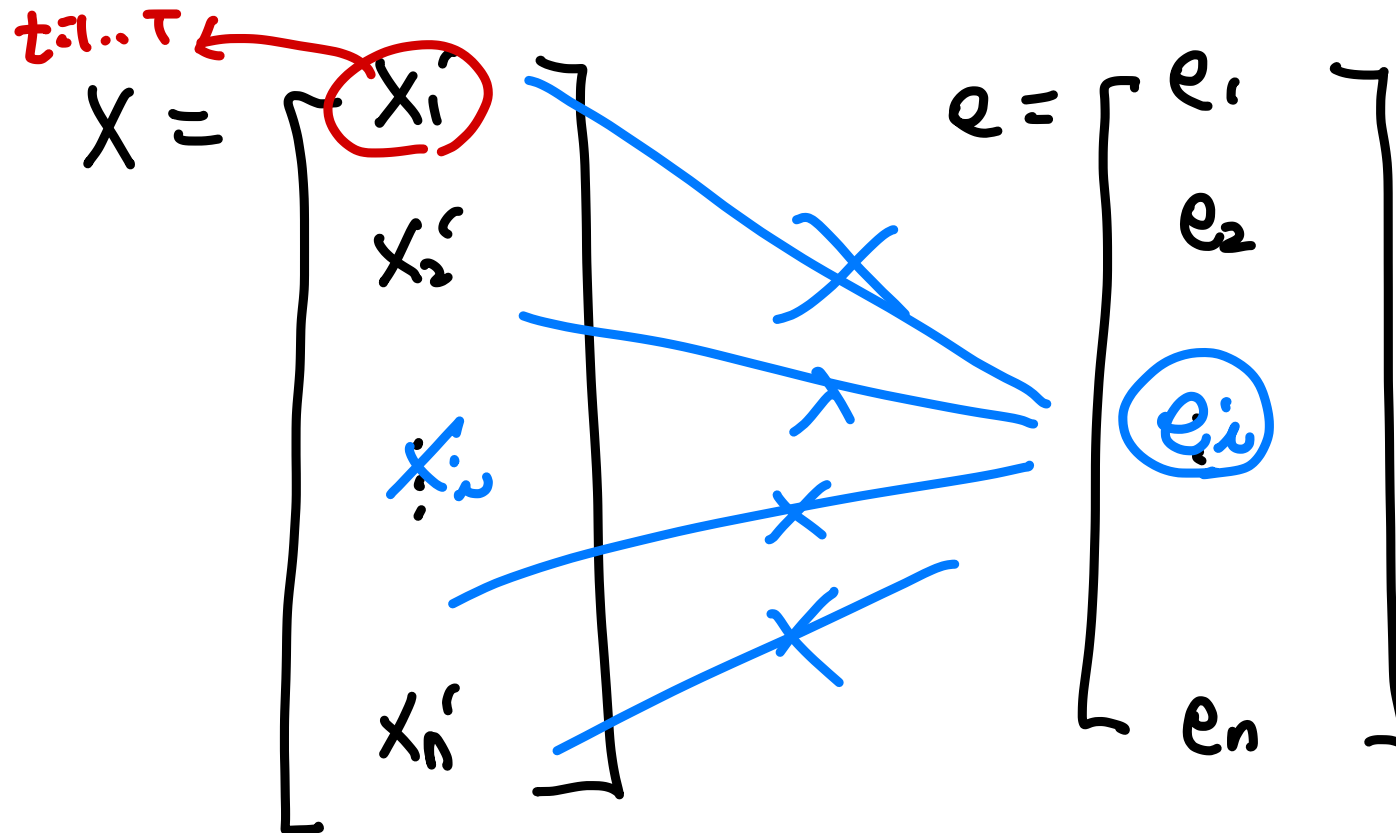
biased!

hint) $n \rightarrow \infty \Rightarrow$
bias $\rightarrow 0$

"Large sample theory"

Assumptions

Assumption TS.3 (graphically) indicates that ...



Unbiasedness of OLS

Theorem (Unbiasedness of OLS for Time Series): Under Assumptions TS.1, TS.2, and TS.3, the OLS estimators are unbiased (conditional on the outcome of the explanatory variables):

$$E(\hat{\beta}_j) = \beta_j, \quad j = 0, \dots, k.$$

- $\{x_{tj}\}$ are allowed to be correlated across time.
- Further, the errors, $\{u_t\}$ are allowed to be correlated across time.
- What we are ruling out with TS.3 is correlation between the errors in any time periods and the explanatory variables in any time period.

TS.3 is too Restrictive.

- The strict exogeneity assumption rules out some practically important situations. First, it can *never* be true when a lagged dependent variable, y_{t-1} , is included among the x_{tj} .
- It imposes some other important restrictions in static and FDL models, too. Consider again

$$y_t = \alpha_0 + \delta_1 z_t + \delta_2 z_{t-1} + \delta_2 z_{t-2} + u_t$$

The Assumption TS.3 requires that u_t is not only uncorrelated with z_t , z_{t-1} , and z_{t-2} , but also all past *and* future outcomes on z .

TS.3 is too Restrictive.

- To ensure u_t is uncorrelated with all past values of z , we need to include enough lags in the FDL model (to take those lags out of the error).
- Unfortunately, there is no way to ensure u_t is uncorrelated with future values, say z_{t+1} . It has to do with why z_t changes over time. Does z_{t+1} react to changes in u_t , that is, to changes in the unobserved part of y_t ?

TS.3 is too Restrictive.

$$E[u_t | \text{all } x_s] \stackrel{?}{=} 0$$

• Suppose we have an FDL relationship between inflation and an interest rate, say, the federal funds rate:

충격
충격: Shock

$$\text{inf}_t = \alpha_0 + \delta_0 \text{ffrate}_t + \delta_1 \text{ffrate}_{t-1} + \delta_2 \text{ffrate}_{t-2} + u_t$$

$u_t > 0$
If we assume two lags of the FF rate suffice, we need not worry about correlation between u_t and further lags of ffrate .

⇒ Fed ↑
• But perhaps a positive shock to inflation at time t – that is, $u_t > 0$ – leads the Fed to increase ffrate the next period. Then ffrate_{t+1} and u_t are correlated, violating strict exogeneity.

Correlation 3m.

TS.3 is too Restrictive.

OLS??

아주 제한적인 상황에서 OLS를 쓸 수
있다! " 동기간 (동시성)간
상관이 없으면 OK,
(X, u) "

- Assumption TS.3 (**strict exogeneity**) is a too strong assumption although it guarantees the unbiasedness of OLS.
- So we need to give up the unbiasedness of OLS most of times for time series analysis.
- Instead, we can take a much weaker assumption called '**contemporaneous exogeneity**'.

~~~~~  
~~~~~  
나랑이 북미악음

Alternative assumption to TS.3

- A weaker (and more desirable) assumption is **contemporaneous exogeneity**,

$$E(u_t | \mathbf{x}_t) = 0$$

which implies uncorrelatedness within each time period,

$$\text{Corr}(x_{tj}, u_t) = 0, j = 1, \dots, k,$$

but does not restrict correlations between the error and explanatory variables across different time periods.

- **With a large sample**, the OLS estimator is consistent under 'contemporaneous exogeneity'. That is, as sample size T increases, the OLS estimator converges to the true values of model parameters.

$$\hat{\beta} = \frac{\sum_{t=1}^T x_t y_t}{\sum_{t=1}^T x_t x_t'} \quad \Leftrightarrow \quad \hat{\beta} = (X'X)^{-1} X'Y$$

$$y_t = x_t' \beta + u_t$$

$$X = \begin{pmatrix} x_1' \\ \vdots \\ x_T' \end{pmatrix} \quad Y = \begin{pmatrix} y_1 \\ \vdots \\ y_T \end{pmatrix}$$

$$\hat{\beta} = \beta + \frac{\frac{1}{n} \sum_{t=1}^T x_t u_t}{\frac{1}{n} \sum_{t=1}^T x_t x_t'}$$

if $E(\hat{\beta}) = \beta$ then
 "unbiased"
 $E(u_t | X) = 0$

<OLS assumptions>

① $\frac{1}{n} \sum_{t=1}^T x_t x_t' \xrightarrow{p} E(x_t x_t')$ $< \infty$ if $n \rightarrow \infty$
 (by L.L.N)

② $\frac{1}{n} \sum_{t=1}^T x_t u_t \xrightarrow{p} E(x_t \cdot u_t) = 0$

by Law of Iterative expectation,
 if $E[u_t | x_t] = 0$,

$$\begin{aligned}
 &E(u_t \cdot x_t) \\
 &= E[E(u_t \cdot x_t) | x_t] \\
 &= E[x_t \cdot E(u_t | x_t)] = 0
 \end{aligned}$$

$\Rightarrow$ OLS
 "consistent"

Assumptions

Assumption TS.4 (Homoskedasticity): For all t ,

$$\text{Var}(u_t|\mathbf{X}) = \sigma^2.$$

- Variance of u_t cannot depend on $\mathbf{x}_t$, $\mathbf{x}_s$, or change over time for reasons we do not know.
- Given our cross section analysis, not surprising some form of homoskedasticity is needed for usual variance formulas and Gauss-Markov Theorem to hold.
- Violation of TS.4 called “heteroskedasticity,” as in the CS case.



Assumptions

Assumption TS.5 (No Serial Correlation): For all $t \neq s$,

$$\text{Corr}(u_t, u_s | \mathbf{X}) = 0$$

- Sometimes called the **no autocorrelation** assumption.
- With cross section (CS) data, we assumed random sampling, which implies that outcomes on all variables are independent across observation. We have not assumed random sampling here.
- In practice, do not worry about the conditioning on $\mathbf{X}$. Just consider $\text{Corr}(u_t, u_s)$.

Assumptions

Assumption TS.6 (Normality): $\{u_t\}$ is independent of $\mathbf{X}$ and is independent and identically distributed (i.i.d.) as $Normal(0, \sigma^2)$:

$$u_t \sim Normal(0, \sigma^2), \quad t = 1, 2, \dots, n$$

- Assumptions TS.1 to TS.6 are the **classical linear model (CLM) assumptions for time series**.
- As in the cross-sectional case, we cannot perform exact inference unless we add a normality assumption.
- Even though TS.6 (Normality) is violated, we can still use all results for confidence interval and test statistics as long as our sample size is large enough. (all results are asymptotically valid.)

Summary

- Linearity (TS.1) + No perfect multicollinearity (TS.2) + Strict exogeneity (TS.3) + Homoskedasticity (TS.4) + no serial correlation (TS.5) + Normality (TS.6)

→ OLS is BLUE (Gauss-Markov Theorem) and the usual OLS variance (and standard error) formulas hold.

→ Exact statical inference is possible.

Summary

- Linearity (TS.1) + No perfect multicollinearity (TS.2) + Strict exogeneity (TS.3) + Homoskedasticity (TS.4) + no serial correlation (TS.5)

→ OLS is BLUE (Gauss-Markov Theorem) and the usual OLS variance (and standard error) formulas hold.

→ Approximate statical inference is possible. We can use all results for confidence interval and test statistics as long as our sample size is large enough

Summary

- Linearity (TS.1) + No perfect multicollinearity (TS.2) + Strict exogeneity (TS.3) + Homoskedasticity (TS.4)

→ OLS is still unbiased. But it is no longer BLUE. (OLS is not the best estimator. GLS is the best estimator.)

→ **The usual OLS variance (and standard error) formulas do not hold. We need to use serial correlation-robust (autocorrelation-robust) standard errors for statistical inference.**

→ Approximate statical inference is possible. We can use all results for confidence interval and test statistics as long as our sample size is large enough

Summary

- Linearity (TS.1) + No perfect multicollinearity (TS.2) + Strict exogeneity (TS.3)

→ OLS is still unbiased. But it is no longer BLUE. (OLS is not the best estimator. GLS is the best estimator.)

→ **The usual OLS variance (and standard error) formulas do not hold. We need to use heteroskedasticity-and-autocorrelation-robust (HAC-robust) standard errors for statistical inference.**

→ Approximate statical inference is possible. We can use all results for confidence interval and test statistics as long as our sample size is large enough

Summary

The usual OLS variance (and standard error) formulas do not hold. We need to use heteroskedasticity-and-autocorrelation-robust (HAC-robust) standard errors for statistical inference:

Summary

```
1 import statsmodels.api as sm
2 import numpy as np
3
4 # Assuming y and x are already defined as your
   dependent and independent variables
5 # For example purposes, let's create some dummy data
6 np.random.seed(0)
7 n = 100
8 x = np.random.rand(n, 1)
9 y = x * 2 + np.random.normal(size=(n, 1))
10
11 # Adding constant to the model for the intercept
12 x = sm.add_constant(x)
13
14 # Fit OLS model with HAC robust standard errors
15 model = sm.OLS(y, x).fit(cov_type='HAC', cov_kwds={
    'maxlags': 1})
16
17 # Print out the summary of the model
18 print(model.summary())
```

Summary

- Linearity (TS.1) + No perfect multicollinearity (TS.2) + **contemporaneous exogeneity** (This is the set of assumptions people usually take in practice).

→ **OLS is biased but is consistent.** OLS is no longer BLUE. (OLS is not the best estimator. GLS is the best estimator.)

→ **The usual OLS variance (and standard error) formulas do not hold. We need to use heteroskedasticity-and-autocorrelation-robust (HAC-robust) standard errors for statistical inference.**

→ Approximate statical inference is possible. We can use all results for confidence interval and test statistics as long as our sample size is large enough

The End. Next, we will learn
ARMA models!